

COMMUNITY ENGAGEMENT PROJECT (CEP)

SKILL BRIDGE

Community Employment and Skill Matching Portal

Submitted in Partial Fulfilment

of the Requirements for the Award of the Degree

of

BACHELOR OF SCIENCE

[To be added: Information Technology / Data Science]

SEMESTER V

By

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Roll Number: 24302A0067

Under the esteemed guidance of

[To be added: Guide name]

[To be added: Guide designation]

Department of Computing

VSIT | Vidyalankar School of
Information Technology

RE-ACCREDITED : GRADE 'A' BY NAAC

DEPARTMENT OF COMPUTING

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

(Autonomous College affiliated to University of Mumbai)

MUMBAI, 400037

MAHARASHTRA

2026 - 2027



VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

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CERTIFICATE

I hereby certify that Gaurav Mhatre (Roll Number: 24302A0067), student of Vidyalankar School of Information Technology, studying in T.Y. B.Sc. [To be added: Information Technology / Data Science], Semester V, has completed a C.E. project titled “Skill Bridge — Community Employment and Skill Matching Portal” at Khalsa College, Matunga, during the academic year 2026–27.

Internal Guide

Head of Department

Department of Computing

Internal Examiner

External Examiner

Date:



Principal

ORGANIZATION VISIT OBSERVATION SHEET

Attach the Organization Visit Observation Sheet duly signed by the reporting officer and mentor.

Organization: Khalsa College, Matunga

Student: Gaurav Mhatre | Roll Number: 24302A0067

[To be added: Signed observation sheet covering all three visits]

PROJECT GUIDE MEETING RECORD

Attach the Project Guide Meeting Record duly signed by the mentor.

Project: Skill Bridge — Community Employment and Skill Matching Portal

Student: Gaurav Mhatre | Roll Number: 24302A0067

[To be added: Signed guide meeting record]

DECLARATION

I, Gaurav Mhatre, student of Vidyalankar School of Information Technology in T.Y. B.Sc. [To be added: Information Technology / Data Science], Semester V, declare that the work described in this report concerns our project “Skill Bridge — Community Employment and Skill Matching Portal”, completed as part of CEP in 2026–27.

The fieldwork consisted of three visits to Khalsa College, Matunga. Our team collected real student responses by manually circulating a Google Form. The technical descriptions in this report concern the shared application developed for the project.

Sources used in preparing the report are identified in the references. Supporting forms and signatures are to be attached in their prescribed locations.

Signature of the Student with Date: _____

Gaurav Mhatre

Roll Number: 24302A0067

ACKNOWLEDGEMENT

I acknowledge the opportunity provided by the University of Mumbai and Vidyalkar School of Information Technology to undertake this CEP project. It helped bring together fieldwork, system design and technical verification.

I thank [To be added: Guide name], our faculty mentor, and Khalsa College, Matunga, for their support of the project process. The students who completed the circulated Google Form made an important contribution to the fieldwork.

I am thankful to my teammates, Dhiraj Tendulkar, Vidhi Mahato and Varun Patil. Our shared work forms the basis of this report, and I value the cooperation involved in developing and reviewing the application.

ABSTRACT

Skill Bridge was developed to connect student placement activities with coordinator follow-up through a shared web application. The project formed part of our CEP work at Khalsa College, Matunga, where all three team visits took place. Student information was gathered manually through a circulated Google Form. The purpose was to support a more connected process for opportunity discovery, preparation, applications and requests for assistance, while keeping responsibilities and access distinct.

The solution contains student profiles, opportunity listings, saved records, applications, workshop participation, feedback and guidance. Coordinators manage the operational queues and record decisions. Administrators handle protected system documents, approved diagnostics and backup restoration. Research information is presented publicly through summaries, with individual respondent records limited to authorized staff. Persistent records allow actions and decisions to be retrieved across sessions.

The technical structure consists of a React and TypeScript frontend, an Express API and a SQLite database. Vite prepares the frontend build, Tailwind CSS provides styling and Recharts supports charts. Matching uses fixed weighted comparisons rather than machine learning. It provides reasons and eligibility cautions so that students can interpret the result instead of treating it as a hiring prediction.

The final recorded verification passed 37 acceptance checks, 97 workflow and security checks, 14 lifecycle checks and 9 production-serving checks. The build and TypeScript checks also passed. This evidence supports the tested local operation of the portal, including production startup and administration. It does not replace field-based user acceptance or demonstrate improved placement rates. The project delivers a practical system and a documented technical baseline. Verified survey analysis, visit evidence and formal acceptance records remain necessary supporting material for the final submission.

TABLE OF CONTENT

CHAPTER 1: INTRODUCTION	1
1.1 Project Background.....	1
1.2 Problem Statement.....	1
1.3 Need for the Project	1
1.4 Objectives	2
1.5 Scope.....	2
1.6 SDG Alignment	2
CHAPTER 2: REQUIREMENT ANALYSIS AND LITERATURE REVIEW	3
2.1 Methodology Adopted	3
2.2 Stakeholder Interactions.....	3
2.3 Domain Overview	4
2.4 Existing Process Analysis	4
2.5 Existing Solutions and Related Work	4
2.6 Technologies Studied.....	4
2.7 Research Gap and Need for Proposed Solution	5
2.8 Challenges Identified	5
CHAPTER 3: PROPOSED SOLUTION	6
3.1 Solution Overview	6
3.2 System Architecture.....	6
3.3 Process Flow Diagram	6
3.4 Module Description	6
3.5 Advantages of Proposed Solution.....	7
CHAPTER 4: SYSTEM DESIGN	8
4.1 System Design Overview.....	8
4.2 Use Case Diagram.....	8
4.3 Workflow Diagram	9
4.4 Module Design.....	9
CHAPTER 5: PROJECT IMPLEMENTATION	11
5.1 Development Methodology	11
5.2 Module-wise Implementation	11
5.3 User Interface Solution	12
5.4 Screenshots of Developed Solution	13
CHAPTER 6: TESTING AND VALIDATION	14
6.1 Test Cases	14

6.2 Test Results	15
6.3 User Acceptance Testing	15
6.4 Feedback from Stakeholders	15
6.5 Performance Evaluation	15
CHAPTER 7: RESULTS AND IMPACT ASSESSMENT	16
7.1 Deliverables Submitted	16
7.2 Results Achieved	16
7.3 Impact on Organization	16
7.4 SDG Contribution	16
CHAPTER 8: CONCLUSION AND FUTURE SCOPE	17
8.1 Conclusion	17
8.2 Key Findings	17
8.3 Recommendations	17
8.4 Future Enhancements	17
REFERENCES	18
Research Papers	18
Books Referred	18
Website	18
ANNEXURES	19
Annexure A: Field Visit Photographs	19
ANNEXURE B: SURVEY QUESTIONNAIRE	22
ANNEXURE C: INTERVIEW QUESTIONS	24
ANNEXURE D: REPORTING OFFICER FEEDBACK FORM	25
ANNEXURE G: SOURCE CODE REPOSITORY	26
ANNEXURE H: USER MANUAL	27

TABLE OF FIGURES

Figure 1: System Architecture of Skill Bridge (placeholder)	6
Figure 2: Process Flow of Skill Bridge (placeholder).....	6
Figure 3: Use Case Diagram of Skill Bridge (placeholder)	8
Figure 4: Application and Workshop Workflow of Skill Bridge (placeholder)	9
Figure 5: Key Components of the Skill Bridge Database (placeholder).....	10
Figure 6: Student Workspace of Skill Bridge (placeholder).....	13
Figure 7: Placement Coordinator Workspace of Skill Bridge (placeholder)	13
Figure 8: Research Summary View of Skill Bridge (placeholder)	13
Figure 9: Visit 1 at Khalsa College, Matunga.....	19
Figure 10: Visit 2 at Khalsa College, Matunga.....	20
Figure 11: Visit 3 at Khalsa College, Matunga.....	21

TABLE OF TABLES

Table 1: Project team	1
Table 2: Fieldwork record.....	3
Table 3: Technologies used	4
Table 4: Functional modules.....	7
Table 5: Actors and access.....	8
Table 6: Database entities	9
Table 7: Matching score components	10
Table 8: Implementation mapping.....	11
Table 9: Representative test cases.....	14
Table 10: Recorded automated results	15
Table 11: Repository contents.....	26

CHAPTER 1: INTRODUCTION

1.1 Project Background

The setting for our project was Khalsa College, Matunga, where the team carried out all three visits. We used a Google Form circulated among students to gather information for the project. The resulting application, Skill Bridge, brings placement-related tasks into a shared system with different access for students and coordinators. I describe both the fieldwork context and the implementation so that the relationship between the community problem and the software remains clear.

Table 1: Project team

Member	Roll number
Dhiraj Tendulkar	24302A0004
Vidhi Mahato	24302A0028
Gaurav Mhatre	24302A0067
Varun Patil	24302D0074

1.2 Problem Statement

The central design problem is keeping opportunity information, student preferences and coordinator decisions connected. When these records are handled separately, a student can struggle to identify a suitable opening or understand an application's progress. The proposed portal gives each participant an appropriate workspace and records the interaction between them. Verified numerical findings from the collected Google Form responses are [To be added].

1.3 Need for the Project

Skill Bridge is needed as a practical link between placement information and the actions taken on it. Searchable listings are useful only when students can act on them and coordinators can respond. Bringing applications, workshops and guidance into the same database provides continuity. The project also provides a setting in which role separation and accountable record keeping can be demonstrated.

1.4 Objectives

The following objectives provide a practical basis for evaluating the work. They focus on completed user actions and protected records rather than unmeasured promises about employment outcomes.

Primary Objectives

- Provide a distinct student workspace and coordinator workspace with server-enforced access.
- Support discovery, saving and application for opportunities with visible status updates.
- Connect workshop participation, feedback and guidance with coordinator follow-up.

Secondary Objectives

- Explain opportunity matching through understandable rules and reasons.
- Preserve related records and reject invalid or unauthorized changes.
- Provide aggregate research reporting while protecting individual responses.
- Maintain repeatable tests and practical operating instructions.

1.5 Scope

The project boundary is the student–coordinator workflow and the administration needed to maintain it. The application supports opportunity records, applications, training participation and support requests, with research reporting alongside these operations. Recruitment decisions outside the portal and long-term employment outcomes are beyond the demonstrated scope. External deployment and institutional acceptance records are [To be added].

1.6 SDG Alignment

The educational aspect of the portal supports SDG 4 by organising training and career guidance. Its employment-support functions are consistent with SDG 8 because they make opportunities and preparation requirements easier to explore. The report treats this as alignment of project objectives, not proof that the software has already improved employment conditions.

CHAPTER 2: REQUIREMENT ANALYSIS AND LITERATURE REVIEW

2.1 Methodology Adopted

We gathered community information through three visits to the same college and a Google Form shared manually with students. The field responses were genuine; the final exported response sheet and its confirmed totals should accompany the report. Development proceeded through implementation, review and correction of specific defects. This method kept the application's technical validation separate from claims about the community's experience.

Table 2: Fieldwork record

Item	Confirmed information or record
Organization	Khalsa College, Matunga
Visits	Three visits, all at the same college
Collection method	Google Form manually circulated among students
Visit dates	[To be added]
Confirmed response count	[To be added]
Google Form link and response export	[To be added]
Verified findings and response summary	[To be added]

2.2 Stakeholder Interactions

Student responses formed the primary field input. Our team worked collectively, so the report does not assign unsupported individual responsibilities or separate datasets to members. Placement staff and administrators are relevant stakeholders because the portal gives them management functions. The identities and recorded comments of consulted staff are [To be added].

2.3 Domain Overview

The portal belongs to the employment-preparation and college-placement domain. It treats a vacancy as more than a title by recording skills, location, eligibility and deadlines. It also recognises that students may need preparation or guidance before applying. This domain model explains why the system includes workshops and support requests alongside opportunities.

2.4 Existing Process Analysis

Our documented starting point is the Google Form collection activity. Survey collection and operational placement management serve different purposes: the first records responses, while the second tracks ongoing actions and decisions. The proposed system addresses the latter through persistent records. A confirmed institutional process diagram is [To be added].

2.5 Existing Solutions and Related Work

The project did not establish a benchmark against a named external placement product. The available comparison concerns function: a form collects responses, whereas the portal connects those information needs with accounts, workflows and records. Published studies and independently assessed alternatives must be cited only after verification; those references are [To be added].

2.6 Technologies Studied

Our application uses a TypeScript-based frontend and server with SQLite persistence. The technology selection suits a college project that needs interactive screens, authenticated requests and related records. The table below identifies each tool's actual role. It does not imply a comparative experiment with frameworks that are absent from the repository.

Table 3: Technologies used

Technology	Role in Skill Bridge
React	Component-based frontend and role-specific pages
TypeScript	Typed models and compile-time checks for frontend and server
Vite	Frontend development server and production bundling
Tailwind CSS	Interface layout and responsive styling

Technology	Role in Skill Bridge
Recharts	Research charts and visual summaries
Node.js and Express	Runtime, API routes, authentication and authorization
SQLite through node:sqlite	Persistent relational records, foreign keys and transactions
tsx and oxlint	TypeScript execution and lint checks
Google Forms	Manual circulation and collection of real student field responses

2.7 Research Gap and Need for Proposed Solution

The gap identified for system design is continuity: information about a student's needs should connect to opportunities and recorded actions. Skill Bridge creates that continuity through profiles, matching explanations, applications and support queues. The project makes a practical contribution through integration; formal novelty against published research remains [To be added].

2.8 Challenges Identified

The project's technical difficulties centred on data integrity and predictable behaviour. Restoring records must preserve relationships, users must receive only permitted data, and production startup must not depend on demo accounts. Documented field difficulties, if any, should be recorded from the visit notes. They are not inferred from problems encountered while programming.

CHAPTER 3: PROPOSED SOLUTION

3.1 Solution Overview

The proposed solution uses one application with distinct access levels. A student's account supports personal actions, while staff accounts support coordination and administration. Shared records connect the interfaces without giving every user the same permissions. This design is central to both the project demonstration and its intended operational use.

3.2 System Architecture

A client-server architecture separates presentation from data operations. React renders the screens, an API service sends requests, and Express performs validation and permission checks. SQLite stores the resulting records. Production serving combines the compiled frontend and API behind the server, while configuration is established before storage is opened.

[To be added]

Figure 1: System Architecture of Skill Bridge (placeholder)

3.3 Process Flow Diagram

The workflow connects discovery with follow-up. After authentication, a student chooses an opportunity or support service. The server checks the request and stores it. A coordinator can then review the record and update its status. Returning to the student workspace shows the latest persisted result.

[To be added]

Figure 2: Process Flow of Skill Bridge (placeholder)

3.4 Module Description

Module boundaries follow the project's user activities. Common session and API services support the screens, but application decisions, workshop enrolments and research records remain distinct entities. This makes each workflow easier to understand and test.

Table 4: Functional modules

Module	Responsibility
Authentication and access	Sessions, login/logout, role checks and account isolation
Student profile and matching	Profile preferences, skills and explained opportunity relevance
Opportunities and applications	Listing discovery, saved openings, submission and review status
Workshops	Session management, capacity, enrolment and attendance
Feedback and guidance	Student requests and coordinator responses
Research	Restricted response records and public aggregate reporting
Study and visit records	Three-visit documentation, action plans and evidence fields
Administration	Protected content, approved diagnostics and backup/restore

3.5 Advantages of Proposed Solution

The proposed system improves organisation by connecting related tasks through shared records. Its access model helps separate personal student work from institutional operations. Matching explanations and visible statuses improve the information available for decisions, though the project does not quantify an increase in employment outcomes.

CHAPTER 4: SYSTEM DESIGN

4.1 System Design Overview

The design separates display state from persistent state. Components render information, shared context coordinates refreshes, and the API performs protected operations. Database relationships link students to applications and workshops. This structure supports account isolation and reliable retrieval after a page refresh.

4.2 Use Case Diagram

The use-case model identifies what each role is allowed to accomplish. Student actions concern the signed-in account, coordinator actions concern placement operations, and administrator actions concern system maintenance. Public research views use aggregates rather than individual respondent records.

Table 5: Actors and access

Actor	Permitted scope
Public visitor	Permitted public pages, opportunity information and aggregate research
Student	Own profile, saved listings, applications, enrolments, feedback and guidance
Placement Coordinator	Operational management, decisions, rosters and restricted research
Administrator	Protected configuration, diagnostics, backups and authorized management

[To be added]

Figure 3: Use Case Diagram of Skill Bridge (placeholder)

4.3 Workflow Diagram

Workflow design accounts for both successful and unsuccessful requests. The server verifies ownership, permission and input before writing. A duplicate or invalid request returns an error, while a successful write returns confirmation. Enrolment uses transactional handling to keep participation and capacity consistent.

[To be added]

Figure 4: Application and Workshop Workflow of Skill Bridge (placeholder)

4.4 Module Design

Module implementation is supported by typed models and shared services. Data changes pass through the API rather than remaining only in browser state. The matching module compares department, preferred role, skills, location and a detected percentage cutoff. Its output is an explanation of fit, not a trained predictive model.

Database records and relationships

Table 6: Database entities

Entity or group	Purpose
users; sessions	Account identities, roles and active authentication sessions
student_profiles	Student-specific skills, interests and academic information
opportunities; applications	Listings linked to student submissions and review decisions
saved_opportunities	Student-specific saved listings
training_sessions; workshop_enrollments	Workshop information and participation relationships
feedback_submissions; guidance_requests	Requests and staff follow-up
survey_responses	Structured responses with provenance and controlled access

Entity or group	Purpose
employer_outreach	Staff-managed employer contact records
system_documents; app_metadata	Content documents and database initialisation state

[To be added]

Figure 5: Key Components of the Skill Bridge Database (placeholder)

Opportunity matching process

Table 7: Matching score components

Component	Points	Implemented rule
Department	40	Awarded when the listing includes the profile department
Preferred role	25	Case-insensitive containment between a preferred role and listing title
Skill overlap	Up to 20	Seven points per overlapping listed skill, capped at 20
Location	10	Preferred location matches, or the listing uses remote work
Academic cutoff	5	Entered percentage meets a detected two-digit percentage cutoff

The score is the sum of the awarded components, bounded between 0 and 100. Skill points are calculated as $\min(20, 7 \times \text{number of matching listing skills})$. A score of 75 or more is labelled High Fit; 50–74 is Moderate Fit; 30–49 is Partial Fit; lower scores are General Opportunity.

Text comparisons use case-insensitive containment, so the result depends on the terms entered in the profile and listing. The eligibility check detects a two-digit percentage cutoff and returns an eligibility caution where appropriate. It is not a complete eligibility parser. Students must review the original criteria even when the fit score is high.

CHAPTER 5: PROJECT IMPLEMENTATION

5.1 Development Methodology

Our development method relied on small, verifiable changes to the existing project. Reviews identified defects, the relevant code was adjusted, and checks were rerun. The work covered both visible functionality and less visible concerns such as database integrity and startup configuration. Disposable databases supported repeatable testing.

5.2 Module-wise Implementation

The frontend modules use shared context and API methods instead of managing unrelated local copies of important records. The backend validates requests and applies the corresponding database operation. Coordinator updates and student actions therefore refer to the same stored entities. Administrative backup and diagnostic functions are kept behind elevated permissions.

Table 8: Implementation mapping

Source area	Implementation role
src/pages	User-facing pages
src/components	Reusable interface components
src/context/AppContext.tsx	Shared session and application state
src/services/apiService.ts	Client API communication
src/utils/matching.ts	Rule-based matching function
server/index.ts	API and frontend-serving entry point
server/db.ts	Schema and database initialisation
server/validation.ts	Input and restoration validators
server/config.ts	Environment and runtime configuration
scripts	Tests, provisioning and backup utilities
README.md; .env.example	Setup instructions and configuration examples

Authentication and data integrity

Passwords are hashed with scrypt and a salt. Session tokens are carried in an HttpOnly cookie, and protected routes verify the session and role. Public registration is restricted to the student role. Database uniqueness constraints help prevent duplicate applications and enrolments. Restore operations validate supported content and relationships before transactional updates.

Research summaries

The reporting process groups responses by the requested filters and computes counts, distributions and rating summaries. Non-rated responses such as “Have not attended” are not treated as numeric satisfaction scores. Public reporting uses aggregate endpoints; individual responses are limited to authorized staff. The actual field-response totals and numerical findings are [To be added].

5.3 User Interface Solution

Interface design emphasises distinct workspaces and visible state. Students can follow their own records without using staff management screens. Coordinators can review and update operational work through dedicated controls. Responsive styles are present, although this report does not claim a completed evaluation on every device or assistive technology.

5.4 Screenshots of Developed Solution

Visual evidence of the application belongs in the figure spaces below. Capture the working version with appropriate test records so that the images demonstrate the implementation without disclosing student data.

(add picture)

Figure 6: Student Workspace of Skill Bridge (placeholder)

(add picture)

Figure 7: Placement Coordinator Workspace of Skill Bridge (placeholder)

(add picture)

Figure 8: Research Summary View of Skill Bridge (placeholder)

CHAPTER 6: TESTING AND VALIDATION

6.1 Test Cases

Tests were selected around the behaviour needed by users and the rules protecting their records. This included account isolation, application decisions, capacity checks and safe restoration. A separate production check examined startup and serving. The cases below connect these checks with expected outcomes.

Table 9: Representative test cases

Area	Scenario	Expected result
Authentication	Invalid credentials and unauthenticated requests	Rejected
Role authorization	Student requests staff-only resources	Rejected
Account isolation	One student requests another student's records	Private records not exposed
Application lifecycle	Submit, reject duplicate, record coordinator decision	Expected records and status retained
Workshop lifecycle	Enrol, reject duplicate, cancel, enforce capacity	Participation rules enforced
Research privacy	Public/student access to raw responses	Blocked; aggregate reporting available
Content and restoration	Malformed documents and invalid relationships	Rejected; valid restoration supported
Diagnostics	Client-supplied SQL and approved query identifiers	Raw SQL blocked; approved queries work
Production lifecycle	Fresh database, restart and admin provisioning	Startup and provisioning checks passed

6.2 Test Results

The final recorded test run reported no failures across the listed suites. Production serving included static assets, direct routes and a provisioned administrator's session. Compilation also passed. The results should be read as evidence for tested cases, not a universal guarantee of correctness or security.

Table 10: Recorded automated results

Suite	Checks	Result
Acceptance	37	Passed
Workflow and security	97	Passed
Production lifecycle	14	Passed
Production serving	9	Passed
Total	157	Passed

6.3 User Acceptance Testing

Acceptance by actual users requires a record of attempted tasks, difficulties and feedback. Those results and signatures are [To be added]. The project's automated acceptance checks are technical tests; they do not establish institutional approval or successful task completion by field participants.

6.4 Feedback from Stakeholders

The report recognises the Google Form collection as genuine fieldwork. The response export, analysed findings and documented staff comments are [To be added]. Feedback is not reconstructed from sample text inside the application or from the software test results.

6.5 Performance Evaluation

Performance evaluation must distinguish a functioning system from a benchmarked one. The tests verify expected responses and protected operations, but they do not report latency distributions or sustained concurrency. A repeatable workload, hardware description and measured results are [To be added].

CHAPTER 7: RESULTS AND IMPACT ASSESSMENT

7.1 Deliverables Submitted

The deliverables comprise the application, source modules, database schema, verification scripts and documentation. The repository address and formal submission details are [To be added]. The table below identifies what each artefact contributes without assuming that a live handover has already been approved.

7.2 Results Achieved

The project achieved the implementation of the planned role-based workflows and a passing automated verification run. Matching returns reasons, staff decisions update records, and restoration preserves tested relationships. The result is a technically demonstrated portal; community-level outcome measures require additional evidence.

7.3 Impact on Organization

The system provides a basis for more organised placement support through records that connect users and actions. Its impact depends on maintained listings, staff participation and student use. Evidence of deployment, sustained adoption and measured benefit is [To be added]. The report therefore distinguishes technical delivery from organisational effect.

7.4 SDG Contribution

The implementation reflects the selected SDGs through training access and employment preparation. Its role is to support students and coordination, not to guarantee employment. Continued monitoring would be needed to connect use of the portal with measurable educational or work-related outcomes.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The completed implementation demonstrates how student activities and coordinator decisions can share a consistent workflow. All three field visits were undertaken at Khalsa College, Matunga, and the survey was collected manually through Google Forms. Technical verification is documented separately from community outcomes. The final report should be accompanied by the original evidence and signed records.

8.2 Key Findings

The work highlights the importance of connecting records without mixing permissions. It also shows why fresh-database testing is necessary alongside compilation. The matching process is useful as an explanation of relevance, but it remains limited by profile quality and the stated rules. Findings from the actual survey are [To be added].

8.3 Recommendations

I recommend a small supervised rollout before wider use. Confirm role access, data persistence and recovery on the actual host, then record student and coordinator feedback. Maintain a clear owner for each operational task. Complete the documentary gaps using original field records rather than estimates.

8.4 Future Enhancements

Further versions may support reminders, configurable matching weights and broader progress reporting. These are potential enhancements rather than completed deliverables. Their value should be assessed through user feedback, and the underlying permission and recovery mechanisms should remain part of regression testing.

REFERENCES

Research Papers

[To be added: Verified research papers actually consulted, formatted in MLA style.]

Books Referred

[To be added: Books actually consulted, with verified MLA bibliographic details.]

Website

[To be added: Websites actually consulted and the topics referred to, with verified URLs and access details.]

Project source material:

Skill Bridge Project Team. Skill Bridge: Community Employment and Skill Matching Portal. Project source code, 2026.

Skill Bridge Project Team. “Project Setup and Verification.” README.md, 2026.

Skill Bridge Project Team. “Opportunity Matching.” src/utis/matching.ts, 2026.

Skill Bridge Project Team. “Survey Questionnaire.” src/data/defaultSurveySeed.ts, 2026.

Skill Bridge Project Team. “Verification Suites.” scripts/verify-acceptance.ts, scripts/test-workflows.ts, scripts/test-production-startup.ts, and scripts/verify-production-serving.ts, 2026.

ANNEXURES

The annexures retain the labels prescribed in the supplied format: A, B, C, D, G and H.

Annexure A: Field Visit Photographs

Visit 1 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 9: Visit 1 at Khalsa College, Matunga

Visit 2 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 10: Visit 2 at Khalsa College, Matunga

Visit 3 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 11: Visit 3 at Khalsa College, Matunga

ANNEXURE B: SURVEY QUESTIONNAIRE

The team used a manually circulated Google Form to gather genuine responses. The following questions are available in the project questionnaire. The original collection form and any version changes are [To be added].

1. Which department or course are you studying in?

Response options: Computer Science (CS); Information Technology (IT); Data Science (DS).

2. What is your current year of study?

Response options: First; Second; Third; Fourth; Other.

3. What job role or career domain are you mainly interested in?

Response options: Open text response.

4. How many campus recruitment drives offered roles relevant to your preferred domain during the current academic year?

Response options: 0; 1–2; 3 or more; Not sure.

5. Is the placement cell currently meeting your placement-related needs?

Response options: Yes; Partly; No; Not enough experience to judge.

6. How often do you receive placement information early enough to prepare and apply?

Response options: Always; Sometimes; Rarely; Never; Not applicable.

7. How useful has the placement training you attended been for your preferred role?

Response options: 1–5, where 1 is Not useful and 5 is Very useful; Have not attended.

8. What is the biggest barrier you face in getting a suitable placement?

Response options: Limited relevant roles; Eligibility restrictions; Skill gaps; Late information; Interview preparation; Location or pay mismatch; Other; No major barrier.

9. Which placement support do you need most urgently?

Response options: Relevant employer connections; Technical training; Aptitude preparation; Resume and interview help; Career guidance; Timely alerts; Internship support; Other.

10. How satisfied are you with overall placement support?

Response options: 1–5, where 1 is Very dissatisfied and 5 is Very satisfied; Not enough experience to judge.

11. Briefly describe your experience with the placement process.

Response options: Open text response.

12. What one change would most improve placement support for you?

Response options: Open text response.

ANNEXURE C: INTERVIEW QUESTIONS

The project includes an interview guide for placement-coordination discussions. The questions below reproduce that guide. Interview dates, respondent identity, consent and recorded answers are [To be added]. Illustrative answers in the software are not reproduced as actual staff statements.

1. How are student preferences and skills collected?
2. How many relevant employers and roles were available by department?
3. What explains differences in opportunities?
4. How are new employers approached?
5. What makes specialist employers difficult to attract?
6. How are notices received and published?
7. How are training needs identified?
8. How is training effectiveness measured?
9. What alternatives exist for students excluded by eligibility criteria?
10. Are duties, location, and pay details checked before publication?
11. What staffing, time, or budget constraints exist?
12. Which concerns are already being addressed?
13. What additional actions are feasible, with owners and timelines?
14. What can Skill Bridge support?
15. How is student feedback reviewed and answered?

ANNEXURE D: REPORTING OFFICER FEEDBACK FORM

Blank form for assessment of the prepared project

Project: Skill Bridge — Community Employment and Skill Matching Portal

Organization: Khalsa College, Matunga

Student: Gaurav Mhatre | Roll Number: 24302A0067

Reporting officer name: _____

Designation: _____

Date of review: _____

Relevance of the project to student/community needs

Usefulness and clarity of the developed solution

Observations on the demonstration

Suggested improvements

Overall assessment and acceptance status

Reporting officer signature: _____

Organization stamp, if required: _____

ANNEXURE G: SOURCE CODE REPOSITORY

Repository link: [To be added]

Submitted version or commit identifier: [To be added]

Contents uploaded and verified by: [To be added]

The following inventory identifies the project contents to include in the submission. It does not confirm that a repository has already been published.

Table 11: Repository contents

Source area	Contents
src/pages	User-facing pages
src/components	Reusable interface components
src/context/AppContext.tsx	Shared session and application state
src/services/apiService.ts	Client API communication
src/utils/matching.ts	Rule-based matching function
server/index.ts	API and frontend-serving entry point
server/db.ts	Schema and database initialisation
server/validation.ts	Input and restoration validators
server/config.ts	Environment and runtime configuration
scripts	Tests, provisioning and backup utilities
README.md; .env.example	Setup instructions and configuration examples

Exclude passwords, private environment files, real student records and local database backups from any public repository. Provide authorised access to sensitive evidence separately.

ANNEXURE H: USER MANUAL

Use this guide to follow the main student and staff activities. A user must sign in with the appropriate account before performing protected actions. The live website address is [To be added].

Student workflow

1. Enter the student workspace and complete the profile fields used for matching.
2. Find a listing through browsing or filtering and inspect its requirements.
3. Check the match reasons without treating the score as a selection guarantee.
4. Submit an application once and follow its recorded status.
5. Review workshop details before enrolling or cancelling participation.
6. Record a feedback or guidance request when needed, then sign out securely.

Coordinator workflow

1. Use coordinator credentials to access the operational dashboard.
2. Keep listings and workshops current, including dates and capacity.
3. Review the application records and provide the relevant decision.
4. Maintain workshop participation and attendance from the available roster.
5. Respond to student support requests and use restricted research views responsibly.
6. Treat an error message as an unsuccessful action until the record is verified.

Administrator workflow

1. Authenticate with the administrator account created for the deployment.
2. Use protected content and diagnostic tools only for necessary maintenance.
3. Take a backup before operations that could affect stored data.
4. Check a restoration result and the linked operational records.
5. Store backups securely and log out after administration.

Local setup and production preparation

Install a Node.js version compatible with the project dependencies and native SQLite support, then run npm install in the project folder.

Use `npm run dev` for local development. The configured API and frontend addresses should be read from the terminal output.

Run `npm run typecheck`, `npm run build` and `npm test` before preparing a release.

For production, set `NODE_ENV=production`, a persistent `DB_PATH` and `CLIENT_URL` matching the actual website origin. Configure HTTPS and the intended reverse proxy.

Provision an administrator with explicit credentials using `npm run admin:provision -- <username> <password>`. Use protected credential handling in the deployment environment.

Run `npm start` after the build. Verify login, authorised changes and direct route refreshes on the actual host.

Use `npm run backup:db` for the database backup utility. Test recovery with a disposable copy before relying on a backup. Do not run demo reset commands against the real database.

If an operation fails

An error should be investigated rather than treated as a successful save. Recheck the record, connection and account permissions. Report reproducible failures to the administrator with the relevant steps, keeping personal information private.